BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA

Order Instituting Rulemaking to Establish Energization Timelines.	Rulemaking 24-01-018 (Filed January 25, 2024)

COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON PHASE 1 SCOPING MEMO ISSUES

Zach Woogen
Interim Executive Director

Albert Tapia
Policy Analyst

VEHICLE-GRID INTEGRATION COUNCIL
10265 Rockingham Dr.
Suite#100-4061
Sacramento, California 95827
Telephone: (510) 665-7811
Email: vgicregulatory@vgicouncil.org

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission ("Commission"), the Vehicle-Grid Integration Council ("VGIC")[1] hereby submits these comments on the *Assigned Commissioner's Scoping Memo and Ruling* ("Scoping Memo") and *Email Ruling Adjusting Phase 1 Schedule and Addressing the April 2, 2024 Joint Investor-Owned Utilities' Motion* issued by Administrative Law Judge ("ALJ") Carolyn Sisto on March 28, 2024 and April 8, 2024, respectively.

I. INTRODUCTION.

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of electric vehicles ("EV") and vehicle-grid integration ("VGI") through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from EV deployments and flexible EV charging and

[1] VGIC member companies and supporters include American Honda Motor Co., Inc., BorgWarner, Fermata Energy, Ford Motor Company, General Motors, Nissan Group of North America, Bidirectional Energy, Customized Energy Solutions, dcbel, Eaton Corporation, Emporia Corp., EnergyHub, EV.Energy, FreeWire Technologies, Inc., GridWiz, Kaluza, Landis + Gyr, Leapfrog Power, Inc., Nuvve Holding Corporation, Peak Power, PowerFlex, Qcells, Stellantis N.V., Sumitomo Electric, The Mobility House, Toyota Motor North America, Inc., WeaveGrid, Hoosier Energy, Peninsula Clean Energy, and Sacramento Municipal Utility District. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (https://www.vgicouncil.org/)

discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid.

VGIC below offers its response to Phase 1 questions identified in the Scoping Memo and provides the following summary of our comments:

- VGIC recommends the Commission direct utilities to track and publish the characteristics and project milestone dates for new and upgraded electrical service requests, including details related to flexible service connection.
- VGIC proposes a partial data reporting template for consideration.
- VGIC recommends utilities ensure customers are provided with flexible service connection options in case of exceptionally lengthy energization timelines.

II. RESPONSE TO PHASE 1 QUESTIONS.

A. Issue 1.f: What information should be tracked to assess improvement in utility timelines for energization after the targets set on or before September 30, 2024, are established?

VGIC recommends the Commission direct utilities to track and publish the characteristics and project milestone dates for each new and upgraded electrical service request. This reporting should include:

i. Anonymized **project identification number, utility, ZIP code.**[2]

ii. the **request type** (i.e., new or upgrade), **project status/current phase**, and **size / kW net nameplate rating requested** at the site, expressed in kW.[3]

a. For upgrade requests (i.e., as opposed to new service requests), **incremental size / kW net nameplate rating requested** should be reported.

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[2] These Project Specifics are currently included in the large IOUs' quarterly Interconnection Timeline Report, as required by D.20-09-035 Ordering Paragraph 22.

[3] *Ibid.*

iii. the **tariff(s)** under which the request is made, including Rule 15, Rule 16, Rule 29, and Rule 45.

a. **whether a Rule 21 application has also been submitted** for that site.

b. if applicable, based on reporting under item (ii)(a) above, the **corresponding project number detailed in quarterly Rule 21 Interconnection Timeline Report** published pursuant to Decision ("D.") 20-09-035 Ordering Paragraph ("OP") 22.[4]

iv. the **days lapsed since the initial request on which each relevant milestone is achieved**. Relevant milestones should be those adopted by the Commission as informed by the utilities' April 22, 2024, *Joint Response to the ALJ's Ruling Requesting Utility Responses* and party opening and reply comments on Phase 1.[5]

The Commission should make a reasonable effort to align the energization timeline reporting requirements adopted in this proceeding with the quarterly Rule 21 Interconnection Timeline Reporting requirements.[6] This would create consistency for developers that seek both energization requests and Rule 21 interconnection requests. The Commission should also consider the merit of charting a clear path toward a joint timeline reporting format for load and

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[4] *Decision Adopting Recommendations From Working Groups Two, Three, And Subgroup.* R.17-07-007. Issued September 30, 2020. https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M347/K953/347953769.PDF

[5] *San Diego Gas & Electric Company (U 902-E), Pacific Gas And Electric Company (U 39 E), And Southern California Edison Company (U 338-E) Response To Administrative Law Judge Ruling Directing Utility Responses To Questions Regarding Energization Timelines*. R.24-01-018. Filed April 22, 2024. https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M530/K252/530252757.PDF

[6] The quarterly Rule 21 Interconnection Timeline Reports include 36 data reporting fields based on the 19 timelines directed in D.20-09-035 OP 22.

generation as customers increasingly develop new sites with both technologies (e.g., vehicle-to-grid interconnection requests at new school bus electrification sites).[7]

v. whether *flexible service connection* was:

a. **proposed by the utility** (i.e., following utility assessment indicating an *exceptionally lengthy* energization timeline, based on an adopted definition of "exceptionally lengthy")

b. **accepted by the customer** (i.e., following utility recommendation)

c. **proposed by the customer** (i.e., following customer/vendor assessment of Integration Capacity Analysis or "ICA")

d. **accepted by the utility** (i.e., following customer request).

Currently, a *standardized process* for enabling flexible service connections has not been adopted by the Commission.[8] One process proposed during the March 22, 2024, VGI Forum hosted by the Energy Division staff is that, for sites with *exceptionally lengthy* forecasted energization timelines, utilities may approach customers with an option to elect a flexible service connection. Another process presented at the March 22, 2024, VGI Forum is to allow customers to proactively propose a flexible service connection based on

[7] Notably, these V2G electric school bus electrification projects are actively supported by the California Energy Commission and California Air Resources Board. See, for example, *GFO-22-612 Electric School Bus Bi-Directional Infrastructure* https://www.energy.ca.gov/solicitations/2023-04/gfo-22-612-electric-school-bus-bi-directional-infrastructure and Work Group #2 to Discuss the SB 114 Grants for Zero-Emission School Buses and Infrastructure https://www.energy.ca.gov/sites/default/files/2024-03/Work_Group_%232_to_Discuss_the_SB_114_Grants_for_Zero-Emission_School_Buses_and_Infrastructure_Presentation_ada.pdf

[8] SCE is launching a Load Control Management Systems (LCMS) pilot. *See* SCE Opening Comments on OIR. R.24-01-018. Page 11.

the customer/developer's assessment of the Integration Capacity Analysis. Note that while these options were presented and discussed at the March 22, 2024, VGI Forum, they are not necessarily exclusive to VGI or EVs and could be applied technology-agnostic to all relevant energization requests. **VGIC recommends the Commission prioritize considering these process elements in either (1) Phase 2 of this proceeding, in close coordination with the High DER Proceeding (R.22-07-005), the Rule 21 Proceeding (R.17-07-007), and the new Transportation Electrification Proceeding (R.23-12-018), or (2) in the High DER Proceeding (R.22-07-005), in close coordination with Phase 2 of this proceeding, the Rule 21 Proceeding (R.17-07-007), and the new TE Proceeding (R.23-12-018).**

Specifically, VGIC recommends the Commission consider *flexible service connection* through four distinct focus areas: **(1) grid data access and enabling tools**, including further development of the ICA maps in R.22-07-005 and the development of dynamic operating envelopes or limited *load* profiles building on Resolution E-5296,[9] **(2) process and customer engagement**, as partially detailed herein, **(3) technical requirements and relevant standards** for enabling technologies, including power control systems, co-located or integrated energy storage, and linear generators, and **(4) timelines data collection and reporting** to understand deployment progress, as partially detailed herein.

vi. if applicable based on reporting of data point (v) above:

[9] Resolution E-5296 Approving Specifics and Process of Limited Generation Profiles. Issued March 22, 2024. https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M527/K981/527981713.PDF

a. the **type of the accepted *flexible service connection*** (e.g., single-value static limit, programmed/locally-implemented operating envelope/288-value limited load profile, centrally-controlled/communications-based dynamic load dispatch / ADMS/DERMS-integrated response)[10]

b. the **type of *flexible service connection* technology** in use (e.g., power control systems,[11] systems that monitor usage at a specific point and take action to reduce usage if power exceeds a programmed watt limit,[12] integrated or co-located stationary energy storage,[13] linear generators,[14] etc.).

This data will be critical to developing a common, objective, and data-backed understanding of progress against California's goal for timely energization of new and upgraded electrical load requests, including the specific energization timelines ultimately adopted by the Commission by September 30, 2024. Moreover, this information can help identify precisely where common bottlenecks persist and how severe they are based on site size to ensure future policy development and investments are reasonable and prudent.

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[10] SCE Advice Letters 5138-E and 5138-E-A, effective January 3, 2024, pursuant to January 16, 2024, Energy Division disposition of Advice Letters, detail both "**localized autonomous LCMS**," in which a site operates autonomously to locally control flexible loads to maintain the power import level to the programmed limit, and "**communications-based LCMS**," in which limits received from SCE via a cloud-based service or directly to the load customer facility by an SCE-approved communications gateway.

[11] See, for example, the UL 3141 Outline for Investigation for Power Control Systems. https://www.shopulstandards.com/ProductDetail.aspx?UniqueKey=45654

[12] SCE LCMS approach offers example LCMS systems including those that reduce charging levels for each charger, trip and disconnect certain devices, and reduce power usage on specific devices

[13] See, for example, solutions from FreeWire Technologies, Inc. https://freewiretech.com/

[14] See, for example, solutions from Mainspring Energy. *Fast-Tracking EV Truck Charging: Prologis Creates Solutions Amid Grid Challenges.* November 30, 2023. https://www.prologis.com/blog/fast-tracking-ev-truck-charging-prologis-creates-solutions-amid-grid-challenges

In the TE Policy and Investments Decision (D.22-11-040), the Commission directs the Energy Division to work with a technical consultant to complete an Automated Load Management ("ALM") study to collect data related to flexible service connection technologies specific to EVs.[15] This effort is a one-off initiative to determine what, if any, additional support may be needed for these EV-specific flexible service connection technologies. Collecting the above data, including items listed in (v) and (vi), will provide critical insights into the value of these technologies on an ongoing basis, helping to answer the following policy questions:

- How many customers are utilities approaching with opportunistic flexible service connection options in circumstances of *exceptionally lengthy* energization timelines?
- How many customers elect flexible service connection pathways (e.g., static single-value limits, programmed operating envelopes/limited load profiles, dynamic load control, etc.) and technologies (e.g., power control systems, integrated energy storage, etc.)?
- What types of flexible service connection pathways and technologies are most common?
- How does the customer election of flexible service pathways and technologies impact energization timelines, on average? How does this differ by pathway and technology type?

VGIC proposes the below energization timelines data reporting template for party and Commission consideration:

[15] *Decision on Transportation Electrification Policy and Investment.* D.22-11-040. Issued November 21, 2022. Page 179. https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M499/K005/499005805.PDF

VGIC Comments Section II.A subsection:	Project Specifics							Flexible Service Connection						Milestones pursuant to Commission direction in R.24-01-018				
	(i) - Project Specifics			(ii) - Project Specifics			(iii) - Tariff	(v)(a)-(b) - Utility-Proposed Process		(v)(c)-(d) - Customer-Proposed Process		(vi) - Type and Tech		(iv) - Milestones				
	Project ID	**Utility**	**ZIP Code**	**Request Type**	**Project Status (Current Phase)**	**Size <u>(kW - Net Nameplate Rating)</u>**	**Tariff**	**Flex Connect Proposed by Utility?**	**Flex Connected Accepted by Customer?**	**Flex Connect Proposed by Customer?**	**Flex Connect Accepted by Utility?**	**Flex Connect Type**	**Flex Connect Tech**	**Date of initial Energization Request**	**Days from submission of initial Energization Request to Applicant Final Submission (AFS)**	**Days from Applicant Final Submission to [milestone 3]**	**Days from [milestone x] to customer service transformer energization**	**<u>Notes</u>**
		PG&E		New Load	APPROVED	100	R15 & R16	Yes	Yes	No	N/A	Single-Value	PCS	1/1/2025	8	10	12	
		SCE		New Load	SUBMITTED	60	R29 & R21	No	N/A	Yes	Yes	288-Value	Storage	45658	12	16	10	[R21 Project ID X]

B. *Issue 3: What procedure(s) for customers should exist to report energization delays for new and upgraded electric service? What additional procedure(s) or improvements should be made for customers on or before September 30, 2024?*

VGIC recommends the following improvement for customers on or before September 30, 2024: **Customers receiving exceptionally lengthy energization timeline estimates should be offered information from their utility on flexible service connection options.** VGIC offers additional details and recommendations for this process below in response to Issue 3.a.

Additionally, VGIC recommends the Commission direct utilities to notify customers when a timeline will not be met or is at risk of not being met. The notification should include the category of delay, the reason for the delay, the new expected deadline, and, if applicable, flexible service connection options.[16]

C. *Issue 3.a: How do utilities currently engage with customers that may have pending or missed deadlines in their energization project requests? How should utilities improve engagement with customers?*

VGIC recommends the Commission direct utilities to ensure customers whose projects will be or are currently at risk of experiencing exceptionally lengthy energization timelines are provided with alternative options to proceed, including flexible service connections. Allowing for these system design changes during the review and overall energization process will be critical to

[16] This requirement is already imposed on all Rule 21 interconnection requests. See D.20-09-035 OP 25. https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M347/K953/347953769.PDF

streamlining electrification and grid connection. Customers should also be provided with as much information as feasible to design their projects *before* submitting energization requests, and tools like the Integration Capacity Analysis ("ICA") maps are useful in providing this information.

If utility assessment of the new or upgraded electrical service request results in an exceptionally lengthy timeline, customers should be given the following options to proceed:

(1) wait for necessary upgrades to be completed, at risk of exceeding the targeted energization timelines

(2) withdraw customer request and re-design site (i.e., they go "back to the drawing board") or forego electrification project altogether

(3) elect flexible service connection option(s) based on site characteristics

VGIC recommends that the Commission direct utilities to establish this third option in the Decision to be adopted by September 30, 2024 and that the Commission take additional action to implement related flexible service connection domains as detailed in Section (II)(a)(v) above.

D. Issue 6: What specific criteria should the Commission establish as annual reporting requirements for the electrical corporations pursuant to Pub. Util. Code Section 933.5(a)(2)?

As detailed above in response to Issue 1.f., VGIC recommends the Commission establish clear, robust energization project timeline reporting.

III. <u>CONCLUSION.</u>

VGIC appreciates the opportunity to provide these comments on the Scoping Memo. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

<u>/s/ Zach Woogen</u>

Zach Woogen
Senior Policy Manager
VEHICLE-GRID INTEGRATION COUNCIL

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